

1 Multi-Angle Approach

(rotate for better pose)



Problem

Direct robot→object approach often hits PLANNING_ERROR; planner falls back to a distant waypoint, leaving robot >1 m from target.



When to Apply

- navigate_to_pose returns success but robot is >1 m from target
- Log shows PLANNING_ERROR messages
- Object is near walls, furniture, or room boundaries



Strategy

Try 5 approach angles: direct, $\pm 90^\circ$, $\pm 45^\circ$. Rotate the approach vector around the object to find gaps through furniture. Keep first success.

3 Waypoint Hopping

(progressive navigation)

4

Exploration-Based Search

(free-space probes)

...

Multi-Angle Approach

```
unit = (obj_pos[:2] - robot_pos[:2]) /  
np.linalg.norm(obj_pos[:2] - robot_pos[:2])
```

```
for angle_offset in [0, np.pi/2, -np.pi/2, np.pi/4, -  
np.pi/4]:  
    cos_off, sin_off = np.cos(angle_offset),  
    np.sin(angle_offset)  
    rotated_unit = np.array([  
        unit[0]*cos_off - unit[1]*sin_off,  
        unit[0]*sin_off + unit[1]*cos_off,  
    ])  
    target_xy = obj_pos[:2] - rotated_unit *  
    approach_dist  
    target_yaw = float(np.arctan2(obj_pos[1]-  
    target_xy[1], obj_pos[0]-target_xy[0]))  
    if navigate_to_pose([target_xy[0], target_xy[1],  
    target_yaw]):  
        break
```



Navigation